



WILLIAM & MARY

Neural Networks and Deep Learning (Fall 2026, DATA 442)

Schedule

Monday and Wednesday 2:00 PM to 3:20 PM, ISC 3342 (8/26 to 12/15)

Instructor

Jonathan Colen

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Office TBD

Office hours: TBD or by appointment

Course Description

This course teaches the foundation of Neural Networks and Deep Learning. Students entering into this course should have, at minimum, a background in data preprocessing, cleaning, manipulation, and dimensionality reduction within Python. Through an applied learning project, you will learn how to implement a machine learning project from design to implementation in the context of neural networks. Topics we will cover include the basic building blocks of neural networks, RNNs, convolutional networks and computer vision, backpropagation basics and strategies, and more. The course will utilize PyTorch.

This course is based on the book *Deep Learning* by Goodfellow, Bengio, and Courville, which is available at <https://www.deeplearningbook.org/>. Materials are adapted from a previous edition of the course taught by Cristiano Fanelli in 2024.

Prerequisite(s):

Credit Hours: 3

Course Grading Policy

Grade Distribution		Letter Grade Distribution			
Problem Set 1	15%	≥ 93.00	A	73.00 - 76.99	C
Problem Set 2	15%	90.00 - 92.99	A-	70.00 - 72.99	C-
Problem Set 3	15%	87.00 - 89.99	B+	67.00 - 69.99	D+
Problem Set 4	15%	83.00 - 86.99	B	63.00 - 66.99	D
Final Project	40%	80.00 - 82.99	B-	60.00 - 62.99	D-
Total	100%	77.00 - 77.99	C+	≤ 59.99	F

Attendance

This class does not have an attendance policy. However, regular attendance is encouraged as the majority of course content relevant to assignments will be covered in class.

Classroom Behavior

Please remain civil during discussions to promote the open exchange of ideas and foster a culture of open dialogue. Please bear in mind that all students are entitled to their own opinion.

Late / Poor Performance Policy

Assignments will not be accepted late, except in documented circumstances.

Collaboration

All assignments should be done individually. For the final project, you may work in groups of up to three. The expectations for the project scope will increase depending on the number of group members. Each group will submit a report, and for groups of more than one, a short paragraph should be included along with the report that explains the role of each group member.

Communication

TBD

Assignments

There are 4 problem sets you will turn in throughout this course. Each assignment will contribute 15% towards your final grade.

Final Project

The final project will include a report and potentially a poster presentation based on an analysis you conduct within your own domain of study.

Materials, Resources, and Software

This course is based on the book *Deep Learning* by Goodfellow, Bengio, and Courville, which is available at <https://www.deeplearningbook.org/>. Additional readings will be linked in the course outline below.

A SciClone account is needed to access the William and Mary High Performance Computing (HPC) system. Registration can be started at <https://hpc.wm.edu/acctreq/>

This course is computationally intensive. If you intend to work on your own laptop, please ensure it has a dedicated NVIDIA graphics card with CUDA capabilities. If you do not have a laptop with an NVIDIA graphics card, you should consider online services such as Google Colab. Because online services regularly change, I will not be able to provide support specific to those options.

This course will use Python and PyTorch. If you are using a personal machine, you can find installation instructions for PyTorch here: <https://pytorch.org/get-started/locally/>

Academic Integrity

Academic dishonesty is taken very seriously. Make sure to cite all of your work and do not turn in work that is not yours. Cases of academic dishonesty will be evaluated and acted upon in accordance with William and Mary policies, which can be found at <https://www.wm.edu/offices/communityvalues/sarp/>

Use of Artificial Intelligence

TBD

Course outline

The course outline can be found below. The weekly content may change depending on the the pace and progress of classes throughout the semester.

Week	Class Topics	Other
1	Wed, Aug 26: Course Introduction & History of Deep Learning	
2	Mon, Aug 31: Linear Classifiers & Perceptrons Wed, Sep 2: Logistic Regression, Nonlinear Classifiers and (stochastic) Gradient Descent	
3	Mon, Sep 7: No class Wed, Sep 9: Regularization & Optimization Due Fri, Sep 11: PS 0 - Getting Started on HPC (bonus)	Labor Day Holiday
4	Mon, Sep 14: Optimization & Hyperparameter Tuning Wed, Sep 16: Deep Neural Networks & Backpropagation (from scratch)	
5	Mon, Sep 21: Deep Neural Networks & Backpropagation (in PyTorch) Wed, Sep 23: AutoGrad & Computational Graphs	
6	Mon, Sep 28: Image data & Convolutional Layers Wed, Sep 30: Convolutional Neural Networks Due Fri, Oct 2: PS 1 - Building a NN from Scratch	
7	Mon, Oct 5: Convolutional Neural Networks (continued) Wed, Oct 7: Time Series & Recurrent Neural Networks	
8	Mon, Oct 12: Recurrent Neural Networks (continued) Wed, Oct 14: Graph Neural Networks	
9	Mon, Oct 19: Graph Neural networks (continued) Wed, Oct 21: Transformers Due Fri, Oct 23: PS 2 - CNNs and image data	
10	Mon, Oct 26: Transformers (continued) Wed, Oct 28: Unsupervised learning & Autoencoders	
11	Mon, Nov 2: Probabilistic models & Variational Autoencoders Wed, Nov 4: Variational Autoencoders (continued) Due Fri, Nov 6: PS 3 - NNs and time-series data	
12	Mon, Nov 9: Generative Adversarial Networks Wed, Nov 11: Generative Adversarial Networks (continued)	
13	Mon, Nov 16: Contrastive Representation Learning Wed, Nov 18: Contrastive Representation Learning (continued)	
14	Mon, Nov 23: TBD Due Tue, Nov 24: PS 4 - Generative models & representations Wed, Nov 25: No class	Thanksgiving Break
15	Mon, Nov 30: Interpretability & GradCAM (tentative) Wed, Dec 2: Physics-inspired deep learning (tentative)	
Finals	Dec 7-15: Final Project Report and Presentation	Exact Dates TBD